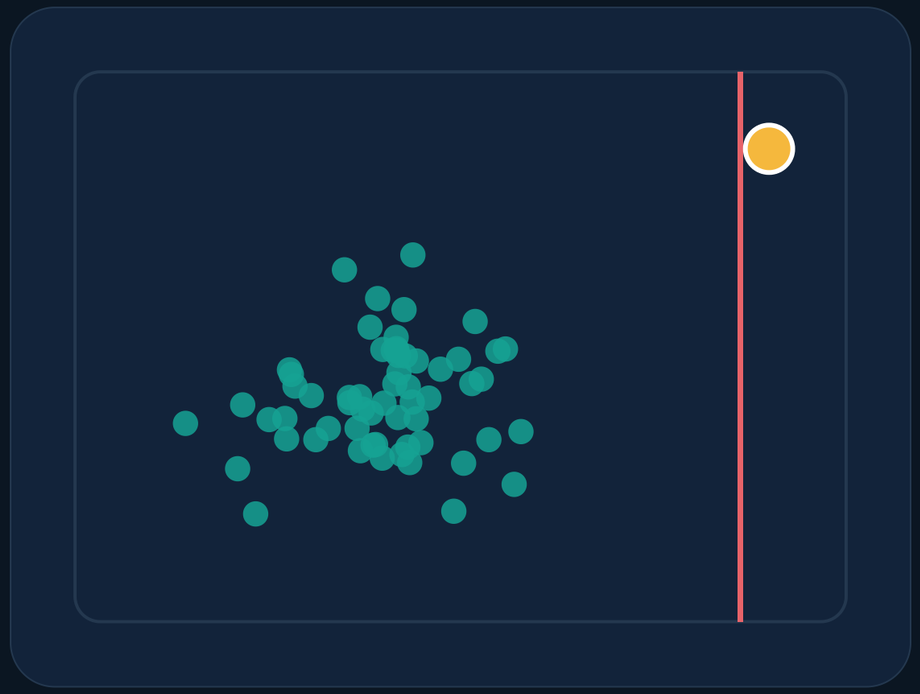
WEEK 4 · DAY 2

Meet scikit-learn + Isolation Forest

Your first machine-learning detector.
Watch it look at six channels at once. 🌲🌲🌲

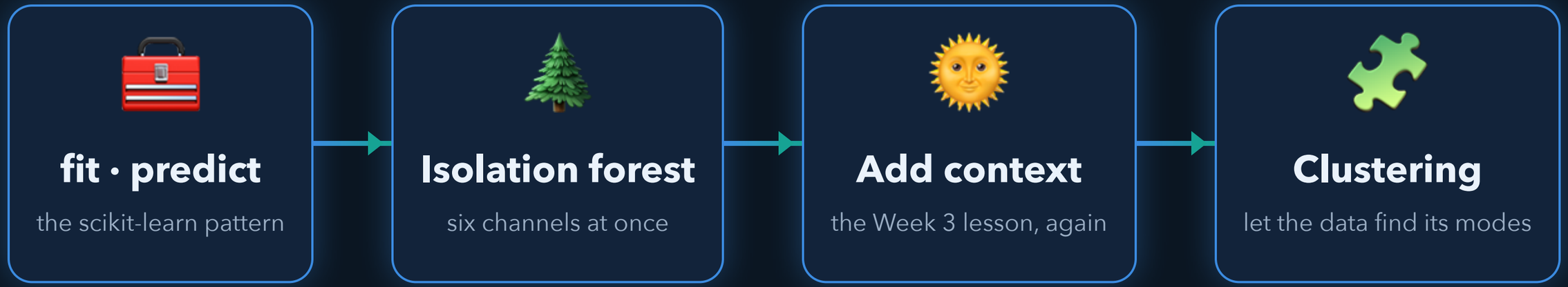


NIGCOMSAT
AFRICAN ROOTED GLOBALLY POSITIONED

Publica Academy

TODAY'S MISSION · 2 HOURS

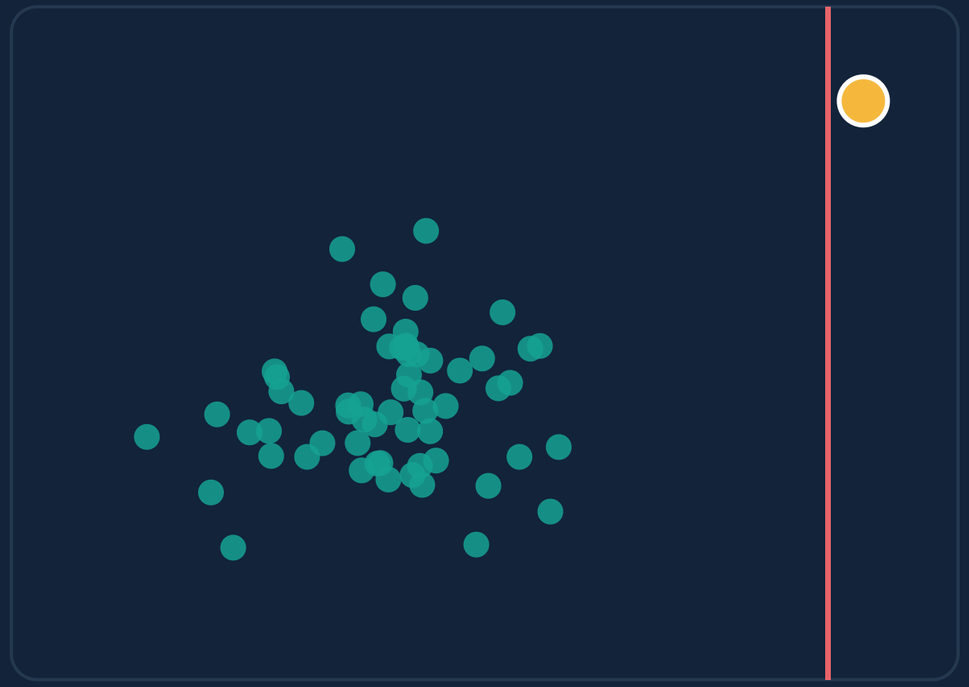
Three verbs, one forest, one honest look



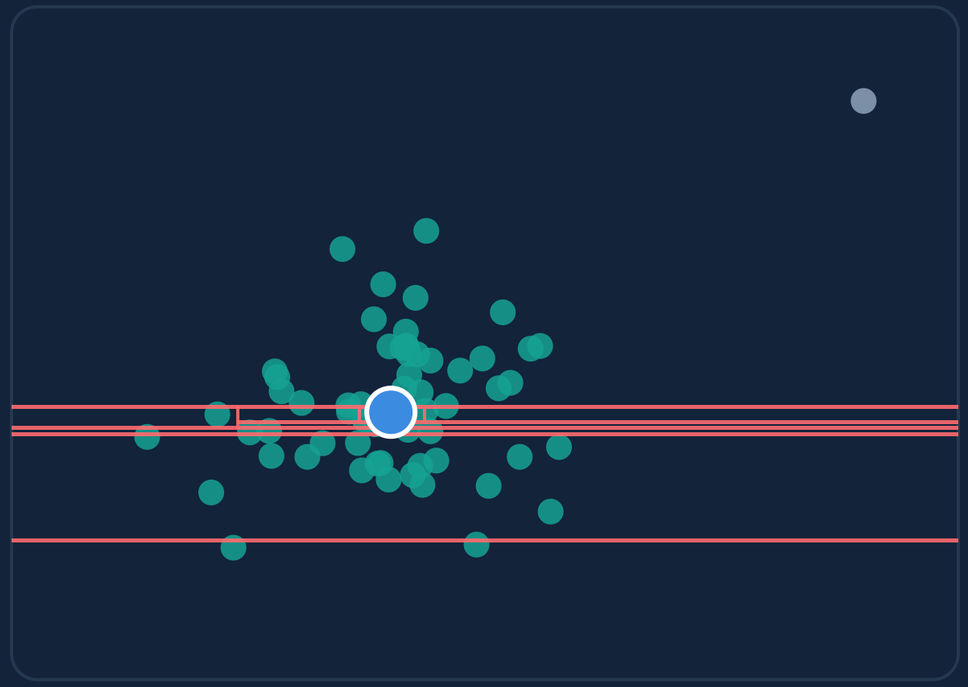
⏱️ 10 min hook · 🎤 30 min slides · 👩🏾‍💻 60 min hands-on · 🏁 20 min wrap-up

QUICK GAME 🏖️

Spot the man in agbada at a beach party



🟡 The man in agbada: **1 cut** and he's alone.



🔵 One guest in shorts: **9 cuts** to get him alone.

THE TWIST

The lifeguard in red shorts also stands out 🛟



The man in agbada

Rare **and** out of place. Somebody should ask him why he's here.

rare + wrong = a fault



The lifeguard

Just as rare, just as easy to isolate. But he **belongs** there. Every evening.

rare + expected = eclipse 🌘

A forest finds what is **rare**. Rare is **not** the same as broken.

WHAT MACHINE LEARNING ADDS

Week 3 watched one channel. This looks at all six at once

array A	0.00 → 29.75
bus V	45.93 → 50.46
SoC %	77.30 → 100.00
batt °C	17.95 → 22.79
TWTA °C	59.15 → 67.00
drive %	69.90 → 80.70



One score per hour

"How easy is this **combination** to isolate?"

It can notice when the **relationship** between channels breaks, even if each one looks fine alone.

PART 1

scikit-learn in 3 verbs 🧰

Learn it once. Every model in the library works the same way.

STEP 1 · BUILD X

Rows are hours. Columns are channels.

hour	array A	bus V	SoC %	batt °C	TWTA °C	drive %
01 Jan 00:00	27.70	50.44	100.00	18.04	59.20	70.10
01 Jan 01:00	27.76	50.46	100.00	18.11	59.53	71.20
01 Jan 02:00	27.85	50.43	100.00	18.14	59.47	71.30
01 Jan 03:00	27.84	50.36	100.00	18.00	59.70	71.10

… 8,760 rows in all

X = bus[CHANNELS]

8,760 hours × 6 channels

STEPS 2-4 · THE PATTERN

Make it. Fit it. Ask it.

```
from sklearn.ensemble import IsolationForest

model = IsolationForest(n_estimators=200, contamination=0.005, random_state=42)
model.fit(X[train])                    # learn what normal looks like (Jan–Aug)
model.predict(X)                       # -1 = odd, 1 = looks normal
model.decision_function(X)             # a score: LOWER = more odd
```

🌲 n_estimators

how many random trees vote

🎚️ contamination

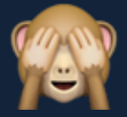
the share of hours you allow it to flag

🎲 random_state

same random cuts every run

BEFORE YOU TRUST IT: TWO CHECKS

It never saw the answer key. So check the count.



Unsupervised

No labels in `fit()`. It only learns what is common.

≈ 44

0.005 × 8,760 hours: what you should expect

58

what it flagged (30 in the training months) ✅

Asked for 0.5% and got 4,000 flags? **Something is wrong in the code.** Stop and check.

PART 2

What did the forest find? 🔍

Six raw channels. Fit on January–August. Look at the whole year.

THE FIRST FOREST · 6 RAW CHANNELS · ALL HOURS

58 flags. All 58 at 21:00. Eclipse. 🌘



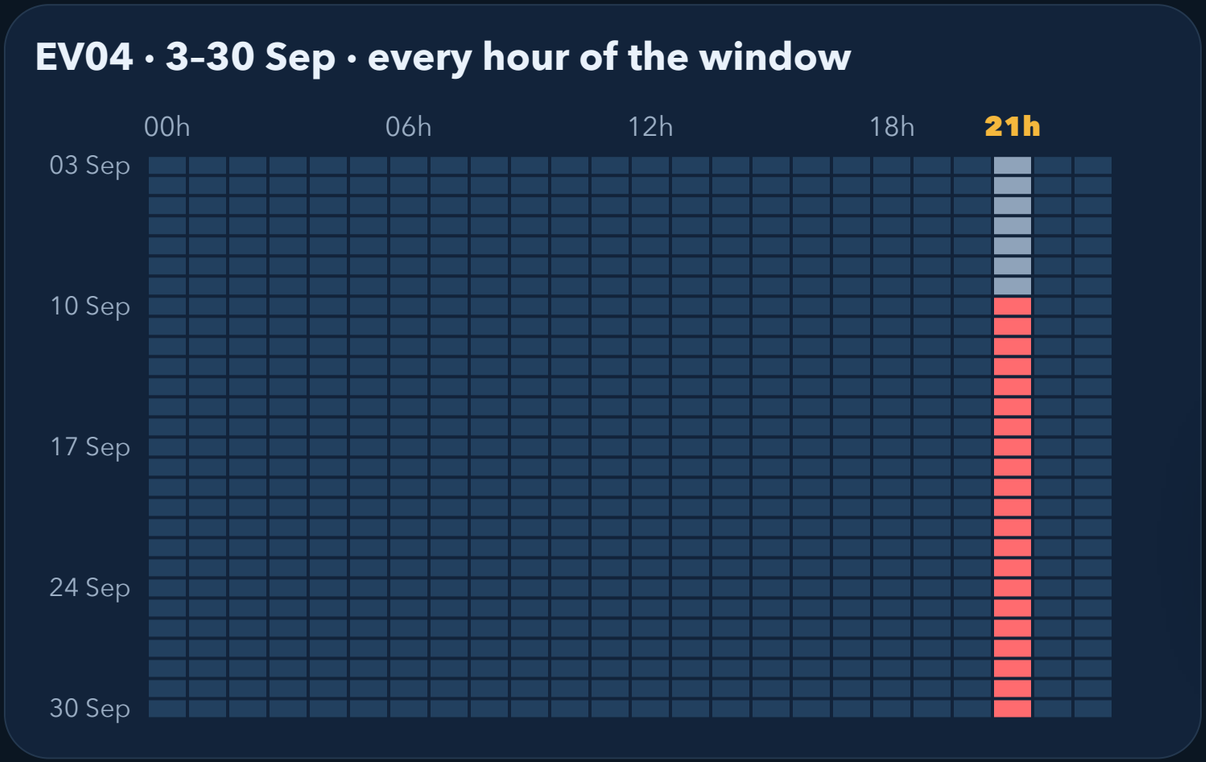
The 10 “oddest” hours of the year:

14 Mar · 21:00 **18 Mar · 21:00** **15 Mar · 21:00** **13 Mar · 21:00** **17 Mar · 21:00** **16 Mar · 21:00** **22 Mar · 21:00**

21 Mar · 21:00 **23 Mar · 21:00** **18 Sep · 21:00**

SCORE IT WITH THE WEEK 4 SCORER

Caught 1 of 5… by luck



21 of 672 hours flagged, every one at **21:00**

First flag **189 hours** after the fault began

EV04 is an **amplifier** fault. The forest was looking at **eclipse**.

37 false alarms outside the events. Grey = eclipse hour · red = flagged.

PART 3

Give it context 🌞

Week 3, Day 1 all over again: condition on the known cause.

THREE STATES, THREE NORMALS

Only judge steady sunlight




```
def steady_sun(bus):
    # no eclipse in this hour or the 3 before it
    return bus["eclipse_flag"].rolling(4).max() == 0
```

FOREST · 6 RAW CHANNELS · STEADY SUN ONLY

4 of 5 caught, 16 false alarms

✅ **EV01**
point
1 of 1 h lit

✅ **EV02**
correlation break
12 of 12 h lit

❌ **EV03**
point
a 12 A drop, missed!

✅ **EV04**
thermal decoupling
1 of 672 h · 25 days late

✅ **EV05**
correlation break
24 of 24 h lit

📅 All 16 false alarms in **January**: the lightest load and coolest amplifier it had seen. The edge of its experience.

🎲 EV04 here is one hour on 28 Sep. With random_state 0 or 1 it's gone completely.

SAME HOURS · SAME SCORER · THE OLD TOOL

The Week 3 chart is still a good tool



Forest, steady sun

caught **4/5** · **16** false alarms

misses EV03, the 12 A drop



3σ chart per channel, steady sun

caught **4/5** · **3** false alarms

EV03 is 34σ low: impossible to miss

For a single-channel spike, the simple chart can be the **better** tool. 💡

Bonus surprise: bus voltage alone gives EV02 and EV05 away. Week 5 Day 1 comes back to this.

PART 4

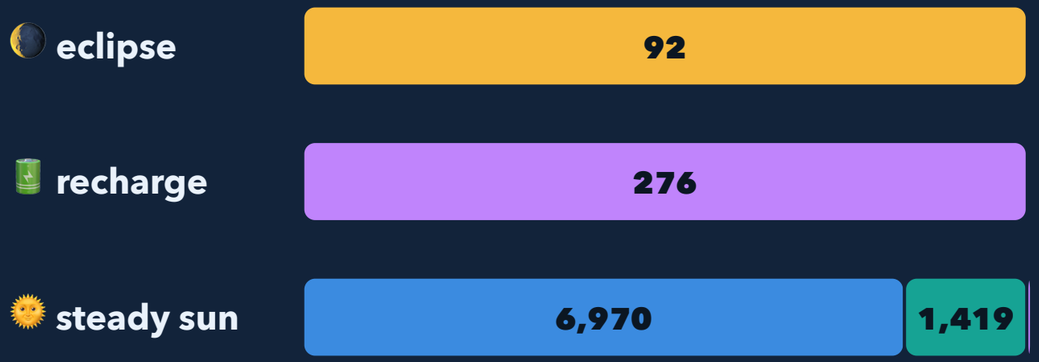
Clustering 🧩

Don't tell it the states. Let it find them.

KMEANS(4) · NO LABELS, NO ECLIPSE_FLAG

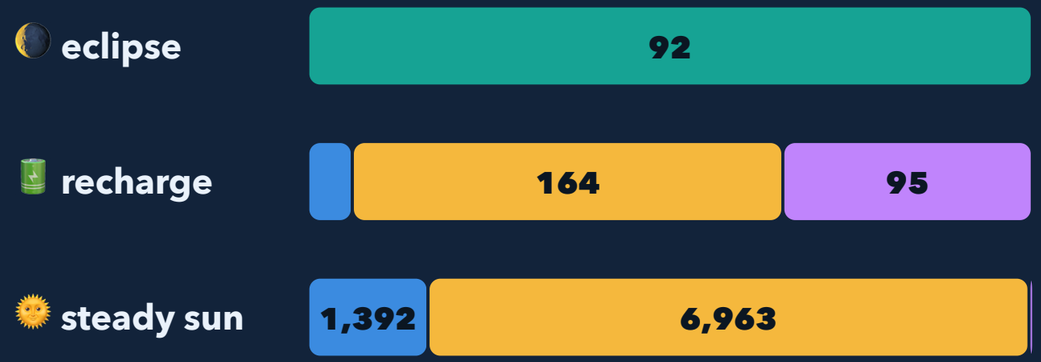
It found eclipse and recharge by itself

✅ Scaled first (StandardScaler)



eclipse → one cluster (92/92) · recharge → one cluster (276/276)

❌ Not scaled



recharge smeared over 3 clusters (164 / 95 / 17)

SO WHEN DOES SCALING MATTER?

Distances care about units. Cuts don't.



k-means: scale it

It measures distance. In steady sun bus voltage wobbles ~0.04 V; array current spans 30 A. Unscaled, the big numbers shout loudest.



Forest: makes no difference

It cuts between min and max of each column, whatever the units. Scores with and without scaling agree: **r = 0.99999999**

```
# a pipeline: scaler + model travel together
km = make_pipeline(StandardScaler(), KMeans(4, n_init=10, random_state=42))
```

"THE ALGORITHM SAYS SO" IS NOT AN ANSWER

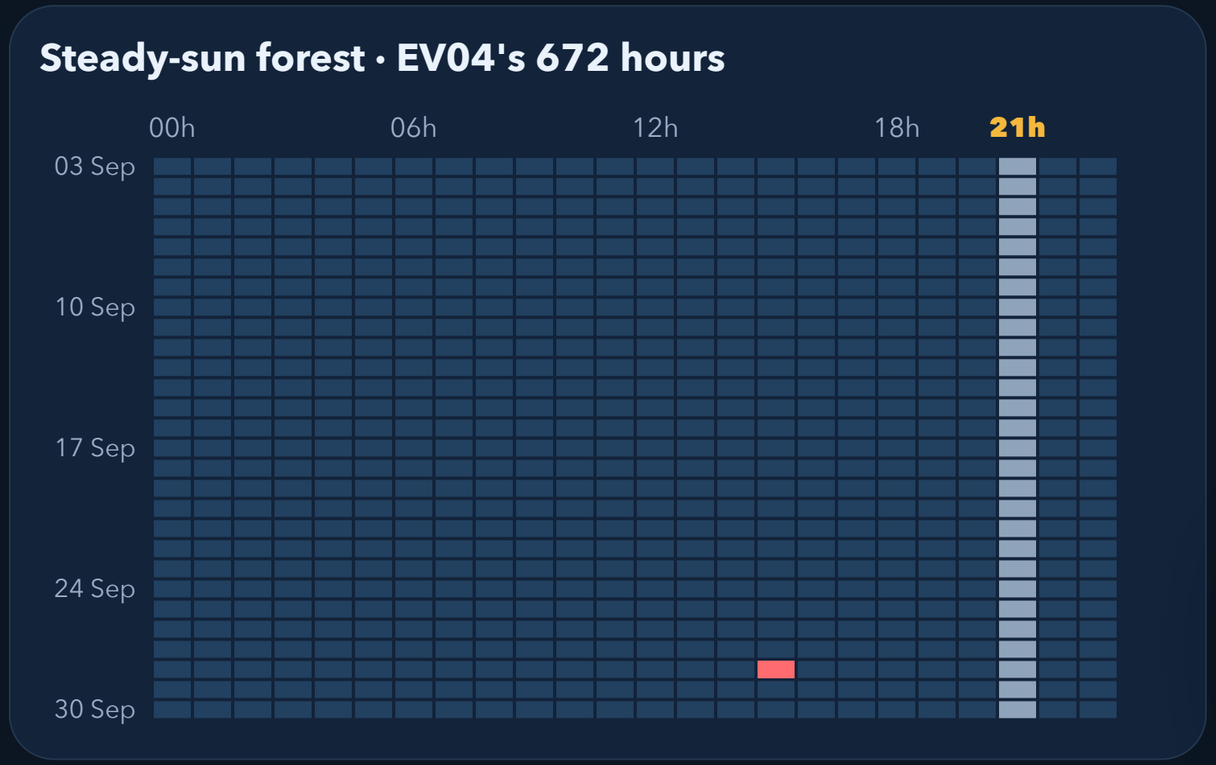
Explain every flag with z-scores

	array A	bus V	SoC %	batt °C	TWTA °C	drive %
EV02 • charge controller 02 May 01:00	−2.3	−6.8	**−20.7**	+1.5	+0.5	+1.2
EV01 • array dropout 19 Feb 11:00	**−33.8**	+1.4	+0.0	+24.8	−1.2	−0.6
January false alarm 01 Jan 10:00	−0.4	+1.7	+0.0	+0.9	−2.1	**−3.1**

z = (value − normal) ÷ normal spread, from steady Jan–Aug hours. The **gold** number is your sentence for the duty engineer.

THE ONE IT CAN'T SEE

EV04: hot for its workload, for four weeks



Amplifier peak **67.0 °C**, its 3σ ceiling **68.0 °C**: looks fine alone

But the RF drive didn't go up. It jumped **~3 °C**: hot for the work it's doing.

The forest saw **1** hour. It has no column for "hot for its workload".

YOUR TURN · 60 MINUTES

Train your first model

1. Open **W4D2-exercise**, save a copy, run STEP 0

A. X, fit, predict: your first forest

B. What did it find? Score it, spot the luck

C. Steady sun: forest vs the Week 3 chart

D. k-means finds the modes; explain a flag



15 checks to pass

🌶️ Pepper task: turn the contamination dial until only the surest faults remain

Keep the **rule book** open next to you.

TODAY IN 3 LINES

What you now know

🧰 **fit • predict • score.** X = hours × channels. One pattern for every scikit-learn model.

🛟 **Rare is not broken.** Give the model context (steady sun) and check **why** it flagged.

📊 **Old tools still win** some fights: the chart caught the spike the forest missed.

Next time 👉 **Give the model eyes**: one new column makes EV04 light up. 👀